

5(a)	any four from: 1 geographical isolation / allopatric (sub)speciation ; 2 little / no, (inter)breeding / gene flow, between <u>populations</u> ; 3 different, <u>selection</u> / <u>selective</u>, pressures (on each population) ; 4 different mutations (in each population) ; 5 <u>adaptation</u> to, environment / habitat / climate / food / vegetation ; 6 gives, morphological / ecological / behavioural, differences ;	4
5(b)(i)	any two from: 1 natural / <u>selection</u> ; 2 dark, colour / fur, selected for / adaptive / good, in woods / warm / shade ; 3 camouflage / right colour, protects against, (named) predators ;	2
5(b)(ii)	any two from: 1 genetic drift / not natural selection ; 2 as small size in north not, adaptive / beneficial ; 3 (small) lose heat faster / harder to keep warm ; 4 smaller fat reserves not good with unstable food supply ; 5 less food in north reduces, growth / size ; 6 AVP ;	2
5(c)	any three from: 1 (hybrid populations have) more, genetic variation / alleles ; 2 genes / mutations / alleles, from both (sub)species ; 3 more potential to adapt / can adapt better (in future) ; 4 genes / alleles, for migration may let some find new habitats ; 5 have, genes / alleles, for, warm and cold / different temperatures ;	3

Question	Answer	Marks
5(d)	any three from: 1 obtain, blood / tissue / DNA ; 2 (test) both (sub)species / migratory and non-migratory ; 3 use PCR to amplify DNA ; 4 use, (gel) electrophoresis / DNA profiling / DNA fingerprinting, and EITHER compare results OR to see genetic, differences / changes ; 5 sequence DNA and, count / identify, differences / changes ; 6 check / test, for correlation between, DNA / genetic profiles / alleles / markers, and migratory behaviour ; 7 use, databases / bioinformatics, to find, name / role of, genes / sequences (that, differ / are associated with migration) ; 8 AVP ;	3

Question	Answer	Marks
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